Data Representation

Michael Cao

Topic 1: Number System

Michael Cao

Learning Objective

In this lesson, you will learn:

- Number system of binary and hexadecimal

Number System

Binary number is the foundation in computer science:

- Binary is used because switches only have ‘on’ and ‘off’ states
- Hexadecimal is used because it’s easier for human to read
- The key for *transformation* is to focus on the weight
- Special attention for the definition of ‘prefix’

Activity: explore the using scenario of hexadecimal number

Decimal prefix name	Symbol used	Factor applied to the value
kilo	k	10^3
mega	M	10^6
giga	G	10^9
tera	T	10^{12}

Binary prefix name	Symbol used	Factor applied to the value
kibi	Ki	2^{10}
mebi	Mi	2^{20}
gibi	Gi	2^{30}
tebi	Ti	2^{40}

Calculation

We use two's complement for addition:

- If we use original number, there will cause error
- Multiple ways to calculate the two's complement
- The philosophy is to find a number that have the same result with overflow
- -128 – 127 is the range of two's complement
- Explore how to transfer a negative two's complement to denary number directly

Practice for deriving the two's complement, as well as addition calculation

We use logical shift as a way of multiple and division

Topic 2: Media and Compression

Michael Cao

ASCII and Unicode

We need a coding scheme to store text in a computer:

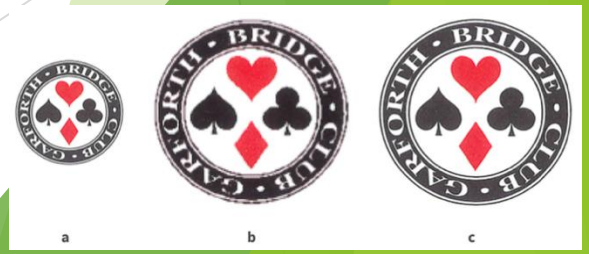
- ASCII: American Standard Code for Information Interchange
  - No need to remember each code, but know the rules
  - Limited characters with 1 byte length (Extended)
- Unicode: Cover more characters with 2 bytes

Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
32	20	<SPACE>	64	40	@	96	60	`
33	21	!	65	41	A	97	61	a
34	22	“	66	42	B	98	62	b
35	23	#	67	43	C	99	63	c
36	24	$	68	44	D	100	64	d
37	25	%	69	45	E	101	65	e
38	26	&	70	46	F	102	66	f
39	27	‘	71	47	G	103	67	g
40	28	(	72	48	H	104	68	h
41	29	)	73	49	I	105	69	i
42	2A	*	74	4A	J	106	6A	j
43	2B	+	75	4B	K	107	6B	k
44	2C	,	76	4C	L	108	6C	l
45	2D	-	77	4D	M	109	6D	m
46	2E	.	78	4E	N	110	6E	n
47	2F	/	79	4F	O	111	6F	o
48	30	0	80	50	P	112	70	p
49	31	1	81	51	Q	113	71	q
50	32	2	82	52	R	114	72	r
51	33	3	83	53	S	115	73	s
52	34	4	84	54	T	116	74	t
53	35	5	85	55	U	117	75	u
54	36	6	86	56	V	118	76	v
55	37	7	87	57	W	119	77	w
56	38	8	88	58	X	120	78	x
57	39	9	89	59	Y	121	79	y
58	3A	:	90	5A	Z	122	7A	z
59	3B	;	91	5B	[	123	7B	{
60	3C	<	92	5C	\	124	7C	|
61	3D	=	93	5D	]	125	7D	}
62	3E	>	94	5E	^	126	7E	~
63	3F	?	95	5F	_	127	7F	<DELETE>

Bitmap

Bitmap is generally used to store an image:

- Pixel: the smallest component of bitmap. (essential information?)
- Color depth: the number of bits used to represent one pixel
- Resolution: image resolution and screen resolution
- File Header: a set of bytes at the beginning identifies the files and contains information about the coding used



Sound

Sound is a continuous format of media:

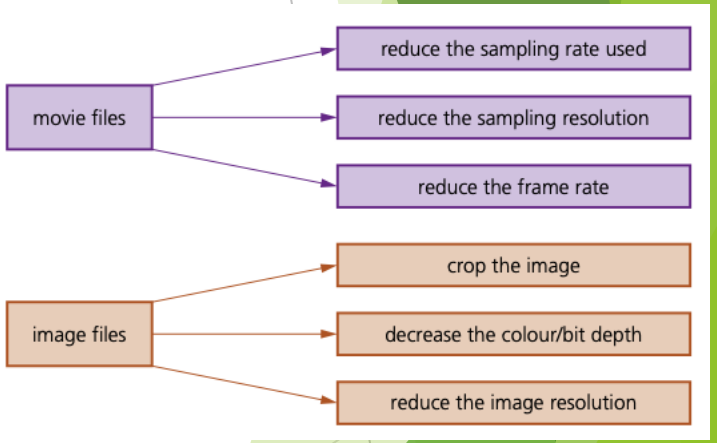
- **ADC and DAC**: convert between analogue and digital data
- **Sampling**: taking measurements at regular intervals and storing the values
- **Sampling resolution**: the number of bits used to store each sample
- **Sampling rate**: the number of samples taken per second

Calculate the file size of images, sound files and videos

Compression

Compression is useful to reduce file size:

- File Size: Calculated by unit and quantity. (also the logic of compression)
- Lossless compression: original file can be restored
  - Coding map table (use code.org to explore)
  - Run-length encoding (consecutive repeating patterns)
- Lossy compression: some data is lost permanently
  - Perceptual shaping
  - Crop or reduce resolution of depth
  - Relative mapping (adjacent frames in video)



Presentation: design / introduce a compression method by your own

Hint: Introduction can include from thesis or existed compression format